HW 0414

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6.
(a)

$$H_0\ \sigma_M^2 = \sigma_F^2$$
$$H_1\ \sigma_M^2 \neq \sigma_F^2$$
$$\hat{\sigma}_M^2 = \frac{SSE_M}{N_M - K} = \frac{97161.9174}{577-4} = \frac{97161.9174}{573} \approx 169.567$$
$$\hat{\sigma}_F = 12.024, \hat{\sigma}_F^2 = 12.024^2 \approx 144.577$$

Using F-statistic:

$$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.567}{144.577} \approx 1.1728$$

Rejection Region:

$$\{f \mid f > F_{0.05/2,577-4,423-4} \approx 1.1968 \text{ or } f < F_{1-0.05/2,577-4,423-4} \approx 0.8377\}$$

Since $0.8377 < 1.1728 < 1.1968$, NOT in the Rejection Region.
Do NOT reject Null Hypothesis H_0 at the 5% significance level.Conclude that there is insufficient evidence to suggest that the random variation in wages differs between men and women.

(b)

$$H_0\ \sigma_{MARRIED}^2 = \sigma_{SINGLE}^2$$
$$H_1\ \sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$$
$$\hat{\sigma}_{SINGLE}^2 = \frac{56231.0382}{400-5} = \frac{56231.0382}{395} \approx 142.357$$
$$\hat{\sigma}_{MARRIED}^2 = \frac{100703.0471}{600-5} = \frac{100703.0471}{595} \approx 169.249$$

Using F-statistic:

$$F = \frac{\hat{\sigma}_{MARRIED}^2}{\hat{\sigma}_{SINGLE}^2} = \frac{169.249}{142.357} \approx 1.1889$$

Rejection Region:

$$\{f \mid f > F_{0.05,600-5,400-5} \approx 1.1647\}$$

Since $1.1889 > 1.1647$, in the Rejection Region.
Reject Null Hypothesis H_0 at the 5% significance level.Conclude that the hypothesis that the variance in wages for married individuals is greater.

(c)
Rejection Region:

$$\{X^2 \mid X^2 > \chi^2_{0.05,4} = 9.488\}$$

Since $NR^2 = 59.03 > 9.488$, in the Rejection Region.
Reject Null Hypothesis H_0 at the 5% significance level.Conclude that heteroskedasticity may exist. While this aligns with the conclusion from (b)—that heteroskedasticity is present—it **cannot** specifically pinpoint whether it is caused solely by 'marital status,' as this test simultaneously accounts for the influence of all variables, such as education and experience.
Since MARRIED is not included as a regressor in (XR8.6b), it is not among the candidates in this NR^2 test. Therefore, this test does not directly provide evidence about whether error variation differs between married and single individuals.

(d)
degrees of freedom:

$$df = 2k + \frac{k(k-1)}{2} = \frac{k^2+3k}{2}$$

$$df = \frac{4^2+3*4}{2} - 2 = 12 \text{ since the } dummy^2 \text{ and } dummy \text{ are same}$$

5%critical value:

$$\chi^2_{0.05,12} = 21.0261$$

Since $NR^2 = 59.03 > 21.0261$, in the Rejection Region.
Reject Null Hypothesis H_0 at the 5% significance level.Conclude that heteroskedasticity may exist.

(e)
Narrower(std bigger): Intercept, EDUC
Wider(std smaller): EXPER, METRO, FEMALE
No, there is no true inconsistency. When there is heterogeneous variance, the OLS standard error is biased. Robust SE is a correction to the true variance. The correction may increase or decrease depending on the correlation structure between the error variance and the explanatory variables.

(f)
Yes, these two results are completely compatible.Since (b) is about the second moment: var(e|x) while (f) is about the first moment: E(WAGE|x).A variable can affect the variability of wages without affecting the average level of wages.

16.
(a)

95% interval estimate:

$$[-135.3298, -28.32302]$$

(b)

Residuals vs INCOME: There is a clear fan-shaped pattern —the spread of residuals is narrow at low income levels and widens substantially as income increases. This is a strong visual indicator of heteroskedasticity, suggesting variance grows with INCOME.
Residuals vs AGE: The spread appears more uniform across AGE values, with no obvious systematic pattern. This suggests AGE is less likely to be the source of heteroskedasticity.
Conclusion: The residual plot strongly suggests heteroskedasticity is present, with error variance increasing with INCOME.

(c)

$$H0 : \sigma^2_{low} = \sigma^2_{high}$$
$$H1 : \sigma^2_{low} < \sigma^2_{high}$$

Goldfeld-Quandt test

Table 1: Goldfeld-Quandt Test for Heteroskedasticity

	Goldfeld-Quandt Test
GQ Statistic	3.1041
df1 (numerator)	86
df2 (denominator)	86
p-value	0.000000

The p-value = $1.64 \times 10^{-7} < 5\%$, so we reject H under 5% significance level.Conclude that there is very strong statistical evidence that the error variance increases with INCOME.

(d)
95% interval estimate:

$$[-139.323, -24.32986]$$

The Robust Interval in (d) is **wider** than the OLS Interval in (a) (given that higher-income individuals exhibit greater variance).

(e)

$$\sigma_i^2 = \sigma^2 \cdot \text{INCOME}_i^2 \Rightarrow w_i = \frac{1}{\text{INCOME}_i^2}$$

$$\text{since Var}\left(\frac{e_i}{\text{INCOME}_i}\right) = \frac{1}{\text{INCOME}_i^2} \cdot (\sigma^2 \cdot \text{INCOME}_i^2) = \sigma^2$$

GLS 95% interval estimate:

$$[-119.8945, -33.71808]$$

GLS Robust 95% interval estimate:

$$[-121.4134, -32.19919]$$

Both the GLS robust intervals and the GLS intervals are narrower than the intervals obtained in (a) and (d), indicating higher precision.